第四章　積分

4.1 不定積分

定義 (反導函數)**.** 給定 $F(x)$，若 $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$，則稱 $F(x)$ 為 $f(x)$ 的反導函數（antiderivative）。

性質. 若 $F(x)$，$G(x)$ 分別為 $f(x)$，$g(x)$ 的反導函數，$c \in \mathbb{R}$。則

- $F(x) + c$ 為 $f(x)$ 的反導函數。
- $c\,F(x)$ 為 $c\,f(x)$ 的反導函數。
- $F(x) + G(x)$ 為 $f(x) + g(x)$ 的反導函數。

結論.

- $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x) \implies \mathrm{d}F(x) = f(x)\cdot\mathrm{d}x \implies F(x) = \displaystyle\int f(x)\cdot\mathrm{d}x = \int f(x)\,\mathrm{d}x$
- $F(x)$ 為 $f(x)$ 的反導函數 $\iff$ $f(x)$ 的反導函數為 $F(x)$ $\iff$ $F(x)$ 的導函數為 $f(x)$ $\iff$ $F(x)$（對 x）的微分為 $f(x)$ $\iff$ $f(x)$（對 x）的（不定）積分為 $F(x)$
- $f(x)$ 的反導函數 $\equiv f(x)$（對 x）的（不定）積分
- 基礎積分集：以下 $\alpha \neq -1$，$a \in \mathbb{R}\setminus\{0\}$。

$f(x)$	x^α	$\dfrac{1}{x}$	e^{ax}	$\sin ax$	$\cos ax$	$\dfrac{1}{\sqrt{a^2-x^2}}$	$\dfrac{1}{a^2+x^2}$
$\displaystyle\int f(x)\,\mathrm{d}x$	$\dfrac{1}{\alpha+1}x^{\alpha+1}$	$\ln\lvert x\rvert$	$\dfrac{1}{a}e^{ax}$	$-\dfrac{1}{a}\cos ax$	$\dfrac{1}{a}\sin ax$	$\sin^{-1}\dfrac{x}{\lvert a\rvert}$	$\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$

- (Liouville) e^{-x^2}，$\dfrac{\sin x}{x}$，x^x 無（初等函數形式之）反導函數！

例.

1. $\displaystyle\int \sqrt{x}\,\mathrm{d}x = \frac{2}{3}x^{\frac{3}{2}} + c$
2. $\displaystyle\int x^\pi\,\mathrm{d}x = \frac{1}{\pi+1}x^{\pi+1} + c$
3. $\displaystyle\int \sin\pi x\,\mathrm{d}x = -\frac{1}{\pi}\cos\pi x + c$
4. $\displaystyle\int \cos xy\,\mathrm{d}x = \frac{1}{y}\sin xy + c$
5. $\displaystyle\int e^{-2x}\,\mathrm{d}x = -\frac{1}{2}e^{-2x} + c$
6. $\displaystyle\int \frac{1}{\sqrt{\pi-x^2}}\,\mathrm{d}x = \sin^{-1}\frac{x}{\sqrt{\pi}} + c$
7. $\displaystyle\int \frac{1}{e+u^2}\,\mathrm{d}u = \frac{1}{\sqrt{e}}\tan^{-1}\frac{u}{\sqrt{e}} + c$
8. $\displaystyle\int \left(\frac{\pi}{x} - e^{\pi x}\right)\mathrm{d}u = \pi\ln x - \frac{e^{\pi x}}{\pi} + c$
9. $\displaystyle\int \frac{2+x^2}{1+x^2}\,\mathrm{d}u = x + 2\tan^{-1}x + c$

變數變換法

結論. $\dfrac{\mathrm{d}}{\mathrm{d}x}f(g(x)) = f'(g(x))\cdot g'(x) \implies \mathrm{d}f(g(x)) = f'(g(x))\cdot g'(x)\,\mathrm{d}x \implies f(g(x)) = \displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x$。令 $u = g(x)$，則 $\dfrac{\mathrm{d}}{\mathrm{d}x}u = g'(x) \implies \mathrm{d}u = g'(x)\,\mathrm{d}x$；故 $\displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x = \int f'(u)\,\mathrm{d}u = f(u) + c = f(g(x)) + c$。

例. 求 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x$。

解. 令 $u = x^2+1$，則 $\mathrm{d}u = 2\,x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\mathrm{d}u$。故 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x = \int \frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int \frac{1}{u}\cdot\frac{1}{2}\,\mathrm{d}u = \frac{1}{2}\ln\lvert u\rvert + c = \frac{1}{2}\ln(x^2+1) + c$。

例. 求 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x$。

解. 令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$。故 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x=\int \sin(3\ln x)\cdot\frac{1}{x}\,\mathrm{d}x=\int \sin 3u\cdot\mathrm{d}u=-\frac{1}{3}\cos 3u+c=-\frac{1}{3}\cos(3\ln x)+c$。

例. 求 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x$。

解. 令 $u=1+e^x$, 則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x=\int \sqrt{1+e^x}\cdot e^x\,\mathrm{d}x=\int \sqrt{u}\cdot\mathrm{d}u=\frac{u^{\frac{3}{2}}}{\frac{3}{2}}+c=\frac{2}{3}(1+e^x)^{\frac{3}{2}}+c$

例. 求 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x$。

解. 令 $u=e^x$，則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x=\int \frac{1}{\sqrt{2-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{2}}+c=\sin^{-1}\frac{e^x}{\sqrt{2}}+c$。

例. 求 $\displaystyle\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x$。

解. $\displaystyle\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x=\int \frac{1}{e^x\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x$。令 $u=e^{-x}$，則 $\mathrm{d}u=-e^{-x}\,\mathrm{d}x \implies e^{-x}\,\mathrm{d}x=-\mathrm{d}u$；故 $\displaystyle\int \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int \frac{1}{\sqrt{1-e^{-2x}}}\cdot e^{-x}\,\mathrm{d}x=-\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=-\sin^{-1}u+c=-\sin^{-1}e^{-x}+c$。

例. 求 $\displaystyle\int \frac{1}{x^2+4x+5}\,\mathrm{d}x$。

解. $\displaystyle\int \frac{1}{x^2+4x+5}\,\mathrm{d}x=\int \frac{1}{(x+2)^2+1}\,\mathrm{d}x$。令 $u=x+2$, 則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int \frac{1}{(x+2)^2+1}\,\mathrm{d}x=\int \frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}(x+2)+c$。

例. 求 $\displaystyle\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x$

解. $\displaystyle\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x=\int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$。令 $u=x-1$，則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x=\int \frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{5}}+c=\sin^{-1}\frac{x-1}{\sqrt{5}}+c$。

例. 求 $\displaystyle\int \tan x\,\mathrm{d}x$。

解. $\displaystyle\int \tan x\,\mathrm{d}x=\int \frac{\sin x}{\cos x}\,\mathrm{d}x$。令 $u=\cos x$，則 $\mathrm{d}u=-\sin x\,\mathrm{d}x$；故 $\displaystyle\int \frac{\sin x}{\cos x}\,\mathrm{d}x=\int \frac{-1}{u}\,\mathrm{d}u=-\ln|u|+c=-\ln|\cos x|+c=\ln|\sec x|+c$

例. 求 $\displaystyle\int \sec x\,\mathrm{d}x$。

解. $\displaystyle\int \sec x\,\mathrm{d}x=\int \frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x$。令 $u=\sec x+\tan x$，則 $\mathrm{d}u=(\sec^2 x+\sec x\tan x)\,\mathrm{d}x$；故 $\displaystyle\int \frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x=\int \frac{1}{u}\,\mathrm{d}u=\ln|u|+c=\ln|\sec x+\tan x|+c$

例. 求 $\int \cos^5 ax\,\mathrm{d}x$。

解. $\int \cos^5 ax\,\mathrm{d}x = \int (1-\sin^2 ax)^2 \cos ax\,\mathrm{d}x$。令 $u=\sin ax$, 則 $\mathrm{d}u = a\cos ax\,\mathrm{d}x$; 故 $\int (1-\sin^2 ax)^2\cos ax\,\mathrm{d}x = \frac{1}{a}\int (1-u^2)^2\,\mathrm{d}u = \frac{1}{a}\int (1-2u^2+u^4)\,\mathrm{d}u = \frac{1}{a}\left(u-\frac{2}{3}u^3+\frac{1}{5}u^5\right)+c = \frac{1}{a}\left(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax\right)+c$。

例. 求 $\int \cos^4 ax\,\mathrm{d}x$。

解. $\int \cos^4 ax\,\mathrm{d}x = \int \left(\frac{1+\cos 2ax}{2}\right)^2 \mathrm{d}x = \frac{1}{4}\int \left(1+2\cos 2ax+\cos^2 2ax\right)\mathrm{d}x = \frac{1}{4}\int \left(1+2\cos 2ax+\frac{1+\cos 4ax}{2}\right)\mathrm{d}x$
$= \int \left(\frac{3}{8}+\frac{1}{2}\cos 2ax+\frac{1}{8}\cos 4ax\right)\mathrm{d}x = \frac{3}{8}x+\frac{1}{4a}\sin 2ax+\frac{1}{32a}\sin 4ax + c$

習題. 以變數變換法求下列不定積分。注意：可能會因為常數項而跟此處答案不同。

1. $\int e^{2x}\sin e^{2x}\,\mathrm{d}x = -\frac{1}{2}\cos e^{2x}+c$
2. $\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = -e^{-\frac{x^2}{2}}+c$
3. $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = -2\cos\sqrt{x}+c$
4. $\int x^2 2^{x^3+1}\,\mathrm{d}x = \frac{2^{x^3+1}}{3\ln 2}+c$
5. $\int \frac{\ln x}{x}\,\mathrm{d}x = \frac{1}{2}(\ln x)^2+c$
6. $\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \sqrt{x^2+2x+3}+c$
7. $\int \frac{x^2}{2+x^6}\,\mathrm{d}x = \frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$
8. $\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x = \tan^{-1}e^x+c$
9. $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\sqrt{1-x^2}+\sin^{-1}x+c$
10. $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x = \sin^{-1}(\tan x)+c$
11. $\int \sin^4 x\cos^5 x\,\mathrm{d}x = \frac{\sin^5 x}{5}-\frac{2\sin^7 x}{7}+\frac{\sin^9 x}{9}+c$
12. $\int \sin^2 x\cos^2 x\,\mathrm{d}x = \frac{x}{8}-\frac{\sin 4x}{32}+c$
13. $\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x = 3\sin^{\frac{1}{3}} x-\frac{3}{7}\sin^{\frac{7}{3}} x+c$
14. $\int \frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x = \frac{\sin^4(\ln x)}{4}-\frac{\sin^6(\ln x)}{6}+c$
15. $\int \frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x = -\frac{1}{\cos x}+\frac{1}{3\cos^3 x}+c$
16. $\int \cos ax\cos bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)}+\frac{\sin(a+b)x}{2(a+b)}+c$
17. $\int \sin ax\sin bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)}-\frac{\sin(a+b)x}{2(a+b)}+c$
18. $\int \sin ax\cos bx\,\mathrm{d}x = -\frac{\cos(a-b)x}{2(a-b)}-\frac{\cos(a+b)x}{2(a+b)}+c$

解.

1. 令 $u=e^{2x}$, 則 $\mathrm{d}u = 2e^{2x}\,\mathrm{d}x \implies e^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$。故 $\int e^{2x}\sin e^{2x}\,\mathrm{d}x = \int \sin e^{2x}\cdot e^{2x}\,\mathrm{d}x = \int \sin u\cdot\frac{1}{2}\mathrm{d}u = -\frac{1}{2}\cos u+c = -\frac{1}{2}\cos e^{2x}+c$。
2. 令 $u=\frac{x^2}{2}$, 則 $\mathrm{d}u = x\,\mathrm{d}x$, 故 $\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = \int e^{-\frac{x^2}{2}}\cdot x\,\mathrm{d}x = \int e^{-u}\,\mathrm{d}u = -e^{-u}+c = -e^{-\frac{x^2}{2}}+c$
3. 令 $u=\sqrt{x}$, 則 $\mathrm{d}u = \frac{1}{2\sqrt{x}}\mathrm{d}x \implies \frac{1}{\sqrt{x}}\mathrm{d}x = 2\,\mathrm{d}u$, 故 $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = \int \sin\sqrt{x}\cdot\frac{1}{\sqrt{x}}\,\mathrm{d}x = \int \sin u\cdot 2\,\mathrm{d}u = -2\cos u+c = -2\cos\sqrt{x}+c$
4. 令 $u=x^3+1$, 則 $\mathrm{d}u = 3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x = \frac{1}{3}\mathrm{d}u$, 故 $\int x^2 2^{x^3+1}\,\mathrm{d}x = \int 2^{x^3+1}\cdot x^2\,\mathrm{d}x = \int 2^u\cdot\frac{1}{3}\mathrm{d}u = \frac{1}{3}\int e^{u\ln 2}\,\mathrm{d}u = \frac{1}{3\ln 2}e^{u\ln 2}+c = \frac{2^{x^3+1}}{3\ln 2}+c$

5. 令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$，故 $\displaystyle\int\frac{\ln x}{x}\,\mathrm{d}x=\int\ln x\cdot\frac{1}{x}\,\mathrm{d}x=\int u\,\mathrm{d}u=\frac{1}{2}u^2+c=\frac{1}{2}(\ln x)^2+c$

6. 令 $u=x^2+2x+3$，則 $\mathrm{d}u=(2x+2)\,\mathrm{d}x\implies(x+1)\,\mathrm{d}x=\dfrac{1}{2}\,\mathrm{d}u$，故 $\displaystyle\int\frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x=\int\frac{1}{\sqrt{x^2+2x+3}}\cdot(x+1)\,\mathrm{d}x=\int\frac{1}{\sqrt{u}}\cdot\frac{1}{2}\,\mathrm{d}u=\sqrt{u}+c=\sqrt{x^2+2x+3}+c$

7. 令 $u=x^3$，則 $\mathrm{d}u=3x^2\,\mathrm{d}x\implies x^2\,\mathrm{d}x=\dfrac{1}{3}\,\mathrm{d}u$，故 $\displaystyle\int\frac{x^2}{2+x^6}\,\mathrm{d}x=\int\frac{1}{2+x^6}\cdot x^2\,\mathrm{d}x=\int\frac{1}{2+u^2}\cdot\frac{1}{3}\,\mathrm{d}u=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}}+c=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$

8. $\displaystyle\int\frac{1}{e^x+e^{-x}}\,\mathrm{d}x=\int\frac{e^x}{e^{2x}+1}\,\mathrm{d}x$。令 $u=e^x$，則 $\mathrm{d}u=e^x\,\mathrm{d}x$，故 $\displaystyle\int\frac{e^x}{e^{2x}+1}\,\mathrm{d}x=\int\frac{1}{e^{2x}+1}\cdot e^x\,\mathrm{d}x=\int\frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}e^x+c$。

9. $\displaystyle\int\frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$。令 $u=\sqrt{1-x^2}$，則 $\mathrm{d}u=\dfrac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x\implies\dfrac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=-\mathrm{d}u$,故 $\displaystyle\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=\int-\mathrm{d}u=-u+c=-\sqrt{1-x^2},\int\frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x=-\sqrt{1-x^2}+\sin^{-1}x+c$。

10. 令 $u=\tan x$，則 $\mathrm{d}u=\sec^2x\,\mathrm{d}x$，故 $\displaystyle\int\frac{\sec^2x}{\sqrt{1-\tan^2x}}\,\mathrm{d}x=\int\frac{1}{\sqrt{1-\tan^2x}}\cdot\sec^2x\,\mathrm{d}x=\int\frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=\sin^{-1}u+c=\sin^{-1}(\tan x)+c$

11. $\displaystyle\int\sin^4x\cos^5x\,\mathrm{d}x=\int\sin^4x\cdot\cos^4x\cdot\cos x\,\mathrm{d}x=\int\sin^4x\cdot(1-\sin^2x)^2\cdot\cos x\,\mathrm{d}x=\int\big(\sin^4x-2\sin^6x+\sin^8x\big)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$，則 $\mathrm{d}u=\cos x\,\mathrm{d}x$，故 $\displaystyle\int\big(\sin^4x-2\sin^6x+\sin^8x\big)\cdot\cos x\,\mathrm{d}x=\int\big(u^4-2u^6+u^8\big)\,\mathrm{d}u=\frac{u^5}{5}-\frac{2u^7}{7}+\frac{u^9}{9}+c=\frac{\sin^5x}{5}-\frac{2\sin^7x}{7}+\frac{\sin^9x}{9}+c$

12. $\displaystyle\int\sin^2x\cos^2x\,\mathrm{d}x=\int\frac{1-\cos 2x}{2}\cdot\frac{1+\cos 2x}{2}\,\mathrm{d}x=\frac{1}{4}\int\big(1-\cos^2 2x\big)\,\mathrm{d}x=\frac{1}{4}\int\big(1-\frac{1+\cos 4x}{2}\big)\,\mathrm{d}x=\frac{1}{4}\int\big(\frac{1}{2}+\frac{\cos 4x}{2}\big)\,\mathrm{d}x=\frac{x}{8}-\frac{\sin 4x}{32}+c$

13. $\displaystyle\int\sin^{-\frac{2}{3}}x\cos^3x\,\mathrm{d}x=\int\sin^{-\frac{2}{3}}x\cdot\cos^2x\cdot\cos x\,\mathrm{d}x=\int\sin^{-\frac{2}{3}}x\cdot(1-\sin^2x)\cdot\cos x\,\mathrm{d}x=\int\big(\sin^{-\frac{2}{3}}x-\sin^{\frac{4}{3}}x\big)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$, 則 $\mathrm{d}u=\cos x\,\mathrm{d}x$, 故 $\displaystyle\int\big(\sin^{-\frac{2}{3}}x-\sin^{\frac{4}{3}}x\big)\cdot\cos x\,\mathrm{d}x=\int\big(u^{-\frac{2}{3}}-u^{\frac{4}{3}}\big)\,\mathrm{d}u=3u^{\frac{1}{3}}-\frac{3}{7}u^{\frac{7}{3}}+c=3\sin^{\frac{1}{3}}x-\frac{3}{7}\sin^{\frac{7}{3}}x+c$

14. 令 $u=\ln x$,則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$,故 $\displaystyle\int\frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x=\int\sin^3(\ln x)\cos^3(\ln x)\cdot\frac{1}{x}\,\mathrm{d}x=\int\sin^3u\cos^3u\,\mathrm{d}u=\int\sin^3u\cdot\cos^2u\cdot\cos u\,\mathrm{d}u=\int\sin^3u\big(1-\sin^2u\big)\cdot\cos u\,\mathrm{d}u=\int\big(\sin^3u-\sin^5u\big)\cdot\cos u\,\mathrm{d}u$。令 $w=\sin u$，則 $\mathrm{d}w=\cos u\,\mathrm{d}u$，故 $\displaystyle\int\big(\sin^3u-\sin^5u\big)\cdot\cos u\,\mathrm{d}u=\int\big(w^3-w^5\big)\,\mathrm{d}w=\frac{w^4}{4}-\frac{w^6}{6}+c=\frac{\sin^4(\ln x)}{4}-\frac{\sin^6(\ln x)}{6}+c$

15. $\displaystyle\int\frac{\sin^3x}{\cos^4x}\,\mathrm{d}x=\int\frac{\sin^2x\cdot\sin x}{\cos^4x}\,\mathrm{d}x=\int\frac{\sin^2x}{\cos^4x}\cdot\sin x\,\mathrm{d}x=\int\frac{1-\cos^2x}{\cos^4x}\cdot\sin x\,\mathrm{d}x=\int\big(\frac{1}{\cos^4x}-\frac{1}{\cos^2x}\big)\cdot$

$\sin x\,\mathrm{d}x$。令 $u=\cos x$，則 $\mathrm{d}u=-\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x=-\mathrm{d}u$，故 $\int\left(\frac{1}{\cos^4 x}-\frac{1}{\cos^2 x}\right)\cdot\sin x\,\mathrm{d}x=\int\left(\frac{1}{u^4}-\frac{1}{u^2}\right)(-)\,\mathrm{d}u=\int\left(\frac{1}{u^2}-\frac{1}{u^4}\right)\mathrm{d}u=-\frac{1}{u}+\frac{1}{3u^3}+c=-\frac{1}{\cos x}+\frac{1}{3\cos^3 x}+c$

16. 由積化和差公式 $\cos\alpha\cos\beta=\frac{1}{2}\big(\cos(\alpha-\beta)+\cos(\alpha+\beta)\big)$，$\int\cos ax\cos bx\,\mathrm{d}x=\int\frac{1}{2}\big(\cos(ax-bx)+\cos(ax+bx)\big)\,\mathrm{d}x=\int\frac{1}{2}\big(\cos(a-b)x+\cos(a+b)x\big)\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}+\frac{\sin(a+b)x}{2(a+b)}+c$。
17. 由積化和差公式 $\sin\alpha\sin\beta=\frac{1}{2}\big(\cos(\alpha-\beta)-\cos(\alpha+\beta)\big)$，$\int\sin ax\sin bx\,\mathrm{d}x=\int\frac{1}{2}\big(\cos(ax-bx)-\cos(ax+bx)\big)\,\mathrm{d}x=\int\frac{1}{2}\big(\cos(a-b)x-\cos(a+b)x\big)\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}-\frac{\sin(a+b)x}{2(a+b)}+c$。
18. 由積化和差公式 $\sin\alpha\cos\beta=\frac{1}{2}\big(\sin(\alpha-\beta)+\sin(\alpha+\beta)\big)$，$\int\sin ax\cos bx\,\mathrm{d}x=\int\frac{1}{2}\big(\sin(ax-bx)+\sin(ax+bx)\big)\,\mathrm{d}x=\int\frac{1}{2}\big(\sin(a-b)x+\sin(a+b)x\big)\,\mathrm{d}x=-\frac{\cos(a-b)x}{2(a-b)}-\frac{\cos(a+b)x}{2(a+b)}+c$。

部份積分法

結論. $\frac{\mathrm{d}}{\mathrm{d}x}\left(u(x)\,v(x)\right)=u(x)\,\frac{\mathrm{d}v(x)}{\mathrm{d}x}+v(x)\,\frac{\mathrm{d}u(x)}{\mathrm{d}x} \implies u(x)\,v(x)=\int u(x)\,\mathrm{d}v(x)+\int v(x)\,\mathrm{d}u(x)$
$\implies \int u\,\mathrm{d}v=u\,v-\int v\,\mathrm{d}u$。

例. 求 $\int x\,e^x\,\mathrm{d}x$。

解. 令 $u=x$，則 $\mathrm{d}u=\mathrm{d}x$。令 $\mathrm{d}v=e^x\,\mathrm{d}x$，則 $v=e^x$。故 $\int x\,e^x\,\mathrm{d}x=x\cdot e^x-\int e^x\,\mathrm{d}x=xe^x-e^x+c$。

例. 求 $\int\ln x\,\mathrm{d}x$。

解. 令 $u=\ln x$，則 $\mathrm{d}u=\frac{1}{x}\mathrm{d}x$。令 $\mathrm{d}v=\mathrm{d}x$，則 $v=x$。故 $\int\ln x\,\mathrm{d}x=\ln x\cdot x-\int x\cdot\frac{1}{x}\,\mathrm{d}x=x\ln x-x+c$。

例. 求 $\int x\tan^{-1}x\,\mathrm{d}x$。

解. 令 $u=\tan^{-1}x$，則 $\mathrm{d}u=\frac{1}{1+x^2}\mathrm{d}x$。令 $\mathrm{d}v=x\,\mathrm{d}x$，則 $v=\frac{x^2}{2}$。故 $\int x\tan^{-1}x\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^2}{2}-\frac{1}{2}\int\frac{x^2}{1+x^2}\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^2}{2}-\frac{1}{2}\int\left(1-\frac{1}{1+x^2}\right)\mathrm{d}x=\tan^{-1}x\cdot\frac{x^2}{2}-\frac{1}{2}\big(x-\tan^{-1}x\big)+c$。

例. 求 $\int x^2\sin x\,\mathrm{d}x$。

解. 令 $u=x^2$,則 $\mathrm{d}u=2x\,\mathrm{d}x$。令 $\mathrm{d}v=\sin x\,\mathrm{d}x$,則 $v=-\cos x$。故 $\int x^2\,\sin x\,\mathrm{d}x=x^2\cdot(-\cos x)+2\int x\cos x\,\mathrm{d}x$。求 $\int x\cos x\,\mathrm{d}x$, 令 $u=x$, 則 $\mathrm{d}u=\mathrm{d}x$。令 $\mathrm{d}v=\cos x\,\mathrm{d}x$, 則 $v=\sin x$。故 $\int x\cos x\,\mathrm{d}x=x\cdot\sin x-\int\sin x\,\mathrm{d}x=x\cdot\sin x+\cos x$。由上，$\int x^2\sin x\,\mathrm{d}x=-x^2\cos x+2(x\sin x+\cos x)+c$。

例. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$ 與 $\int e^{ax}\sin bx\,\mathrm{d}x$，$a$、$b\in\mathbb{R}\setminus\{0\}$。

解. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$, 令 $u=\cos bx$, 則 $\mathrm{d}u=-b\sin bx\,\mathrm{d}x$。令 $\mathrm{d}v=e^{ax}\mathrm{d}x$, 則 $v=\dfrac{1}{a}e^{ax}$。故 $\int e^{ax}\cos bx\,\mathrm{d}x=\cos bx\cdot\dfrac{1}{a}e^{ax}+\dfrac{b}{a}\int e^{ax}\cdot\sin bx\,\mathrm{d}x$。$(\star)$ 求 $\int e^{ax}\sin bx\,\mathrm{d}x$，令 $u=\sin bx$，則 $\mathrm{d}u=b\cos bx\,\mathrm{d}x$。令 $\mathrm{d}v=e^{ax}\mathrm{d}x$，則 $v=\dfrac{1}{a}e^{ax}$。故 $\int e^{ax}\sin bx\,\mathrm{d}x=\sin bx\cdot\dfrac{1}{a}e^{ax}-\dfrac{b}{a}\int e^{ax}\cdot\cos bx\,\mathrm{d}x$。$(\star\star)$ 令 $X=\int e^{ax}\cos bx\,\mathrm{d}x$，$Y=\int e^{ax}\sin bx\,\mathrm{d}x$，則由 $(\star)$，$(\star\star)$ $X=\cos bx\cdot\dfrac{1}{a}e^{ax}+\dfrac{b}{a}Y$，$Y=\sin bx\cdot\dfrac{1}{a}e^{ax}-\dfrac{b}{a}X$。解 X、Y 得 $X=\int e^{ax}\cos bx\,\mathrm{d}x=\dfrac{e^{ax}(a\cos bx+b\sin bx)}{a^2+b^2}$，$Y=\int e^{ax}\sin bx\,\mathrm{d}x=\dfrac{e^{ax}(a\sin bx-b\cos bx)}{a^2+b^2}$。

列表法

結論.

1. 將要微分函數寫左邊、積分函數寫右邊；左邊連續微分、右邊連續積分
2. 依序左上連右下斜線函數相乘，符號正負相間；最下列直線寫成兩邊函數相乘的積分
3. 將上式所得項全部加總即為所求積分

例. 求 $\int (x+3)\,e^{2x}\,\mathrm{d}x$。

解.

Diff		Int
$x+3$	$+$	e^{2x}
1	$-$	$\frac{1}{2}e^{2x}$
0		$\frac{1}{4}e^{2x}$

$$\int (x+3)\,e^{2x}\,\mathrm{d}x=(x+3)\,\frac{1}{2}e^{2x}-\frac{1}{4}e^{2x}$$

例. 求 $\int (x^2-2x)\,e^{kx}\,\mathrm{d}x$。

解.

Diff		Int
x^2-2x	$+$	e^{kx}
$2(x-1)$	$-$	$\frac{1}{k}e^{kx}$
2	$+$	$\frac{1}{k^2}e^{kx}$
0		$\frac{1}{k^3}e^{kx}$

$$\int (x^2-2x)\,e^{kx}\,\mathrm{d}x=(x^2-2x)\,\frac{1}{k}e^{kx}-2(x-1)\frac{1}{k^2}e^{kx}+2\frac{1}{k^3}e^{kx}$$

例. 求 $\int x^4\,\sin 2x\,\mathrm{d}x$。

解.

Diff		Int
x^4	$+$	$\sin 2x$
$4x^3$	$-$	$-\frac{1}{2}\cos 2x$
$12x^2$	$+$	$-\frac{1}{4}\sin 2x$
$24x$	$-$	$\frac{1}{8}\cos 2x$
24	$+$	$\frac{1}{16}\sin 2x$
0		$-\frac{1}{32}\cos 2x$

$$\begin{aligned}&\int x^4\,\sin 2x\,\mathrm{d}x\\&=-x^4\,\frac{1}{2}\cos 2x+4x^3\,\frac{1}{4}\sin 2x+12x^2\,\frac{1}{8}\cos 2x-24x\,\frac{1}{16}\sin 2x-24\,\frac{1}{32}\cos 2x\\&=\left(-\frac{x^4}{2}+\frac{3x^2}{2}-\frac{3}{4}\right)\cos 2x+\left(x^3-\frac{3x}{2}\right)\sin 2x\end{aligned}$$

例. 求 $\displaystyle\int x^5\,e^{ax}\,\mathrm{d}x$。

解.

Diff		Int
x^5	$+$	e^{ax}
$5x^4$	$-$	$\frac{1}{a}e^{ax}$
$20x^3$	$+$	$\frac{1}{a^2}e^{ax}$
$60x^2$	$-$	$\frac{1}{a^3}e^{ax}$
$120x$	$+$	$\frac{1}{a^4}e^{ax}$
120	$-$	$\frac{1}{a^5}e^{ax}$
0		$\frac{1}{a^6}e^{ax}$

$$\begin{aligned}
&\int x^5\,e^{ax}\,\mathrm{d}x\\
&= x^5\,\frac{1}{a}e^{ax} - 5x^4\,\frac{1}{a^2}e^{ax} + 20x^3\,\frac{1}{a^3}e^{ax} - 60x^2\,\frac{1}{a^4}e^{ax} + 120x\,\frac{1}{a^5}e^{ax} - 120\,\frac{1}{a^6}e^{ax}\\
&= \left(\frac{x^5}{a} - \frac{5x^4}{a^2} + \frac{20x^3}{a^3} - \frac{60x^2}{a^4} + \frac{120x}{a^5} - \frac{120}{a^6}\right)e^{ax}
\end{aligned}$$

例. 求 $\displaystyle\int (\sin^{-1}x)^2\,\mathrm{d}x$。

解.

Diff		Int
$(\sin^{-1}x)^2$	$+$	1
$\frac{2\sin^{-1}x}{\sqrt{1-x^2}}$	$-$	x

Diff		Int
$2\sin^{-1}x$	$+$	$\frac{x}{\sqrt{1-x^2}}$
$\frac{2}{\sqrt{1-x^2}}$	$-$	$-\sqrt{1-x^2}$

$$\begin{aligned}
&\int (\sin^{-1}x)^2\,\mathrm{d}x = (\sin^{-1}x)^2\cdot x - \int \frac{2\sin^{-1}x}{\sqrt{1-x^2}}\cdot x\,\mathrm{d}x\\
&= (\sin^{-1}x)^2\cdot x - \int 2\sin^{-1}x\cdot\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x\\
&= (\sin^{-1}x)^2\cdot x - \left(-2\sin^{-1}x\cdot\sqrt{1-x^2} + \int \frac{2}{\sqrt{1-x^2}}\cdot\sqrt{1-x^2}\,\mathrm{d}x\right)\\
&= (\sin^{-1}x)^2\cdot x + 2\sin^{-1}x\cdot\sqrt{1-x^2} - 2x
\end{aligned}$$

例. 求 $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x$。

解.

Diff		Int
$\cos bx$	$+$	e^{ax}
$-b\sin bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\cos bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax}\cos bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}e^{ax}\sin bx - \frac{b^2}{a^2}\int e^{ax}\cos bx\,\mathrm{d}x\\
&\Longrightarrow \int e^{ax}\cos bx\,\mathrm{d}x = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}
\end{aligned}$$

例. 求 $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x$。

解.

Diff		Int
$\sin bx$	$+$	e^{ax}
$b\cos bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\sin bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax}\sin bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\sin bx - \frac{b}{a^2}e^{ax}\cos bx - \frac{b^2}{a^2}\int e^{ax}\sin bx\,\mathrm{d}x\\
&\Longrightarrow \int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}
\end{aligned}$$

遞迴式

- 定義積分 $I_n = \displaystyle\int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x$, $n\in\mathbb{N}$, $n\geqslant 1$, 則 $I_n = \displaystyle\int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x = \frac{1}{a^2}\int \frac{(x^2+a^2)-x^2}{(x^2+a^2)^n}\,\mathrm{d}x = \frac{1}{a^2}\int \frac{1}{(x^2+a^2)^{n-1}}\,\mathrm{d}x - \frac{1}{a^2}\int \frac{x^2}{(x^2+a^2)^n}\,\mathrm{d}x$

- 考慮 $\frac{1}{a^2}\int \frac{x^2}{(x^2+a^2)^n}\,\mathrm{d}x$, 令 $u=x$, 則 $\mathrm{d}u=\mathrm{d}x$。令 $\mathrm{d}v=\frac{x}{(x^2+a^2)^n}\,\mathrm{d}x$, 則 $v=-\frac{1}{2(n-1)}\frac{1}{(x^2+a^2)^{n-1}}$, 故 $\int \frac{x^2}{(x^2+a^2)^n}\,\mathrm{d}x=-\frac{1}{2(n-1)}\frac{x}{(x^2+a^2)^{n-1}}+\frac{1}{2(n-1)}\int\frac{1}{(x^2+a^2)^{n-1}}\,\mathrm{d}x=-\frac{1}{2(n-1)}\frac{x}{(x^2+a^2)^{n-1}}+\frac{1}{2(n-1)}I_{n-1}$
- 所以 $I_n=\frac{1}{a^2}\int\frac{1}{(x^2+a^2)^{n-1}}\,\mathrm{d}x-\frac{1}{a^2}\int\frac{x^2}{(x^2+a^2)^n}\,\mathrm{d}x=\frac{1}{a^2}I_{n-1}-\frac{1}{a^2}\Big(-\frac{1}{2(n-1)}\frac{x}{(x^2+a^2)^{n-1}}+\frac{1}{2(n-1)}I_{n-1}\Big)=\frac{1}{a^2}\frac{2n-3}{2(n-1)}I_{n-1}+\frac{1}{a^2}\frac{1}{2(n-1)}\frac{x}{(x^2+a^2)^{n-1}}$
- 由此遞迴關係及 $I_1=\frac{1}{a}\tan^{-1}\frac{x}{a}$, 可得 I_n, $n\in\mathbb{N}$, $n>1$。例: 令 $n=2$, 則 $I_2=\frac{1}{2a^2}I_1+\frac{x}{2a^2(x^2+a^2)}=\frac{1}{2a^3}\tan^{-1}\frac{x}{a}+\frac{x}{2a^2(x^2+a^2)}$。

習題. 以部份積分法求下列不定積分。注意: 可能會因為常數項而跟此處答案不同。

1. $\int x(\ln x)^3\,\mathrm{d}x=\frac{x^2}{2}\Big((\ln x)^3-\frac{3(\ln x)^2}{2}+\frac{3\ln x}{2}-\frac{3}{4}\Big)+c$
2. $\int x^2\tan^{-1}x\,\mathrm{d}x=\frac{x^3}{3}\tan^{-1}x-\frac{x^2}{6}+\frac{\ln(1+x^2)}{6}+c$
3. $\int x^5e^{-x^2}\,\mathrm{d}x=-\frac{1}{2}e^{-x^2}(x^4+2x^2+2)+c$
4. $\int xe^{\sqrt{x}}\,\mathrm{d}x=2\,e^{\sqrt{x}}\left(x\sqrt{x}-3x+6\sqrt{x}-6\right)+c$
5. $\int \sin(\ln x)\,\mathrm{d}x=\frac{x\,(\sin(\ln x)-\cos(\ln x))}{2}+c$
6. $\int\frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x=\ln x\cdot 2\sqrt{1+x}-4\sqrt{1+x}-2\ln(\sqrt{1+x}-1)+2\ln(\sqrt{1+x}+1)+c$

解.

1. 令 $u=(\ln x)^3$, 則 $\mathrm{d}u=3(\ln x)^2\cdot\frac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v=x\,\mathrm{d}x$, 則 $v=\frac{x^2}{2}$。故 $\int x(\ln x)^3\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot 3(\ln x)^2\cdot\frac{1}{x}\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\frac{3}{2}\int x(\ln x)^2\,\mathrm{d}x$。令 $u=(\ln x)^2$, 則 $\mathrm{d}u=2\,\ln x\cdot\frac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v=x\,\mathrm{d}x$, 則 $v=\frac{x^2}{2}$。故 $\int x(\ln x)^2\,\mathrm{d}x=(\ln x)^2\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot 2\ln x\cdot\frac{1}{x}\,\mathrm{d}x=(\ln x)^2\cdot\frac{x^2}{2}-\int x\ln x\,\mathrm{d}x$。令 $u=\ln x$, 則 $\mathrm{d}u=\frac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v=x\,\mathrm{d}x$, 則 $v=\frac{x^2}{2}$。故 $\int x\ln x\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\frac{1}{2}\int x\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\frac{x^2}{4}$。以上, $\int x(\ln x)^3\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\frac{3}{2}\Big((\ln x)^2\cdot\frac{x^2}{2}-\big(\ln x\cdot\frac{x^2}{2}-\frac{x^2}{4}\big)\Big)=\frac{x^2}{2}\big((\ln x)^3-\frac{3(\ln x)^2}{2}+\frac{3\ln x}{2}-\frac{3}{4}\big)+c$
2. 令 $u=\tan^{-1}x$, 則 $\mathrm{d}u=\frac{1}{1+x^2}\,\mathrm{d}x$; $\mathrm{d}v=x^2\,\mathrm{d}x$, 則 $v=\frac{x^3}{3}$。故 $\int x^2\tan^{-1}x\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x$。又 $\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\frac{(x^3+x)-x}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\frac{x(x^2+1)-x}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\big(x-\frac{x}{x^2+1}\big)\,\mathrm{d}x$。令 $w=x^2+1$, 則 $\mathrm{d}w=2x\,\mathrm{d}x\implies x\,\mathrm{d}x=\frac{1}{2}\,\mathrm{d}w$, 故 $\int\frac{x}{x^2+1}\,\mathrm{d}x=\int\frac{1}{x^2+1}\cdot x\,\mathrm{d}x=\int\frac{1}{w}\cdot\frac{1}{2}\,\mathrm{d}w=\frac{1}{2}\ln w+c=\frac{1}{2}\ln(x^2+1)+c$。以上, $\int x^2\tan^{-1}x\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\frac{x^2}{6}+\frac{\ln(x^2+1)}{6}+c$。

3. 令 $w = x^2$, 則 $\mathrm{d}w = 2x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\,\mathrm{d}w$, 故 $\displaystyle\int x^5 e^{-x^2}\,\mathrm{d}x = \int e^{-x^2}\cdot(x^2)^2\cdot x\,\mathrm{d}x = \int e^{-w}\cdot w^2\cdot\frac{1}{2}\,\mathrm{d}w = \frac{1}{2}\int w^2\,e^{-w}\,\mathrm{d}w = -\frac{1}{2}\,e^{-w}(w^2+2w+2) = -\frac{1}{2}\,e^{-x^2}(x^4+2x^2+2)+c$。

Diff		Int
w^2	$+$	e^{-w}
$2w$	$-$	$-e^{-w}$
2	$+$	e^{-w}
0		$-e^{-w}$

$$\int w^2 e^{-w}\,\mathrm{d}w = -w^2\,e^{-w} - 2w\,e^{-w} - 2\,e^{-w} = -e^{-w}(w^2+2w+2)$$

4. 令 $w = \sqrt{x}$, 則 $w^2 = x$, $\mathrm{d}x = 2w\,\mathrm{d}w$, 故 $\displaystyle\int xe^{\sqrt{x}}\,\mathrm{d}x = \int w^2 e^w\cdot 2w\,\mathrm{d}w = 2\int w^3 e^w\,\mathrm{d}w = 2\,e^w(w^3 - 3w^2+6w-6) = 2\,e^{\sqrt{x}}\,(x\sqrt{x} - 3x + 6\sqrt{x} - 6) + c$

Diff		Int
w^3	$+$	e^w
$3w^2$	$-$	e^w
$6w$	$+$	e^w
6	$-$	e^w
0		e^w

$$\int w^3\,e^w\,\mathrm{d}w = w^3\,e^w - 3w^2\,e^w + 6w\,e^w - 6\,e^w = e^w(w^3 - 3w^2 + 6w - 6)$$

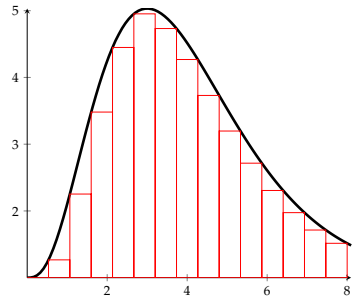
5. 令 $w = \ln x$, 則 $e^w = x$, $\mathrm{d}x = e^w\,\mathrm{d}w$, 故 $\displaystyle\int \sin(\ln x)\,\mathrm{d}x = \int \sin w\cdot e^w\,\mathrm{d}w = \int e^w\,\sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2} = \frac{x\,(\sin(\ln x) - \cos(\ln x))}{2} + c$

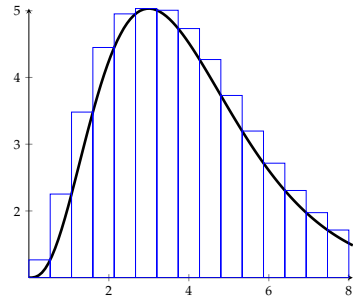
Diff		Int
$\sin w$	$+$	e^w
$\cos w$	$-$	e^w
$-\sin w$	$+$	e^w

$$\int e^w\,\sin w\,\mathrm{d}w = e^w\,\sin w - e^w\,\cos w - \int e^w\,\sin w\,\mathrm{d}w$$

$$\implies \int e^w\,\sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2}$$

6. 令 $u = \ln x$, 則 $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v = \dfrac{1}{\sqrt{1+x}}\,\mathrm{d}x$, 則 $v = 2\sqrt{1+x}$。故 $\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x\cdot 2\sqrt{1+x} - 2\int\frac{\sqrt{1+x}}{x}\,\mathrm{d}x$。令 $w = \sqrt{1+x}$, 則 $w^2 = 1+x \implies x = w^2-1$, $2w\,\mathrm{d}w = \mathrm{d}x$, 故 $\displaystyle\int\frac{\sqrt{1+x}}{x}\,\mathrm{d}x = \int\frac{w}{w^2-1}\cdot 2w\,\mathrm{d}w = 2\int\frac{w^2}{w^2-1}\,\mathrm{d}w = 2\int\frac{(w^2-1)+1}{w^2-1}\,\mathrm{d}w = 2w + 2\int\frac{1}{w^2-1}\,\mathrm{d}w = 2\sqrt{1+x} + 2\int\frac{1}{w^2-1}\,\mathrm{d}w$。由部份分式 $\dfrac{1}{w^2-1} = \dfrac{1}{2}\left(\dfrac{1}{w-1} - \dfrac{1}{w+1}\right)$, $\displaystyle\int\frac{1}{w^2-1}\,\mathrm{d}w = \frac{1}{2}\int\left(\frac{1}{w-1} - \frac{1}{w+1}\right)\mathrm{d}w = \frac{1}{2}\big(\ln(w-1) - \ln(w+1)\big) = \frac{1}{2}\big(\ln(\sqrt{1+x}-1) - \ln(\sqrt{1+x}+1)\big)$。以上, $\displaystyle\int\frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x\cdot 2\sqrt{1+x} - 2\left(2\sqrt{1+x} + 2\left(\frac{1}{2}\big(\ln(\sqrt{1+x}-1) - \ln(\sqrt{1+x}+1)\big)\right)\right) = \ln x\cdot 2\sqrt{1+x} - 4\sqrt{1+x} - 2\ln(\sqrt{1+x}-1) + 2\ln(\sqrt{1+x}+1) + c$。





4.2 定積分

定義. 令 f 在 $[a,b]$ 連續。

- $\mathbb{P}: a = x_0 < x_1 < x_2 < \cdots < x_n = b$ 為 $[a,b]$ 的分割。
- $\Delta x_k = x_k - x_{k-1}$，$\|\mathbb{P}\| = \max\{\,|\Delta x_k| \mid 1 \leqslant k \leqslant n\}$，$x_{k-1} \leqslant \xi_k \leqslant x_k$，$k = 1, 2, \ldots, n$
- $u_k = \sup\{\,f(x) \mid x_{k-1} \leqslant x \leqslant x_k\}$，$l_k = \inf\{\,f(x) \mid x_{k-1} \leqslant x \leqslant x_k\}$，$k = 1, 2, \ldots, n$
- $R(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} f(\xi_k)\,\Delta x_k$，$U(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} u_k\,\Delta x_k$，$L(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} l_k\,\Delta x_k$；
  顯然 $L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。
- $\displaystyle\int_a^b f(x)\,\mathrm{d}x \equiv \lim_{\substack{n\to\infty \\ \|\mathbb{P}\|\to 0}} R(f,\mathbb{P})$；若此極限存在，稱 f 在 $[a,b]$ 可積（分）。

性質 (定積分基本性質)**.** 令 f、g 在包含 a、b、c 之區間為可積，α、$\beta \in \mathbb{R}$。則

1. $\displaystyle\int_a^a f(x)\,\mathrm{d}x = 0$
2. $\displaystyle\int_a^b f(x)\,\mathrm{d}x = -\int_b^a f(x)\,\mathrm{d}x$
3. $\displaystyle\int_a^b \big(\alpha\, f(x) + \beta\, g(x)\big)\,\mathrm{d}x = \alpha \int_a^b f(x)\,\mathrm{d}x + \beta \int_a^b g(x)\,\mathrm{d}x$
4. $\displaystyle\int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x$
5. $\displaystyle\int_a^b f(x)\,\mathrm{d}x \leqslant \int_a^b g(x)\,\mathrm{d}x$，若 $f(x) \leqslant g(x)\ \forall\, x \in [a,b]$，$a \leqslant b$
6. $\displaystyle\left|\int_a^b f(x)\,\mathrm{d}x\right| \leqslant \int_a^b |\,f(x)\,|\,\mathrm{d}x, \quad a \leqslant b$
7. $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 0$，若 $f(x)$ 為奇函數
8. $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x$，若 $f(x)$ 為偶函數

例.

1. $\displaystyle\int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2 - x^2}\,\mathrm{d}x = \pi$（半徑 $\sqrt{2}$ 之半圓面積）
2. $\displaystyle\int_{-1}^{2} \operatorname{sgn}(x)\,\mathrm{d}x = 2 \cdot 1 - 1 \cdot 1 = 1$

3. $\displaystyle\int_{-\pi}^{\pi} \sin(x^7 - 5x^3)\,\mathrm{d}x = 0$ ($\sin(x^7 - 5x^3)$ 為奇函數)

4. $\displaystyle\int_{-6}^{6} e^{-x^4}\sin(\sin x)\,\mathrm{d}x = 0$ ($e^{-x^4}\sin(\sin x)$ 為奇函數)

5. $\displaystyle\int_{-4}^{4} \left(e^x - e^{-x}\right)\mathrm{d}x = 0$ ($e^x - e^{-x}$ 為奇函數)

6. $\displaystyle\int_{-100}^{100} \left(e^{9x^5-2x^7} - e^{-9x^5+2x^7}\right)\mathrm{d}x = 0$ ($e^{9x^5-2x^7} - e^{-9x^5+2x^7}$ 為奇函數)

7. $\displaystyle\int_{-a}^{a} |x|\,\mathrm{d}x = 2\int_0^a |x|\,\mathrm{d}x = a^2$ ($|x|$ 為奇函數；兩 $a \times a$ 直角三角形面積)

8. 定義 $\displaystyle\int_1^x \frac{1}{\tau}\,\mathrm{d}\tau \equiv \ln x$，則 $\displaystyle\int_{\frac{1}{4}}^{3} \frac{1}{x}\,\mathrm{d}x = \int_{\frac{1}{4}}^{1} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = -\int_1^{\frac{1}{4}} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = \ln 12$

9. $\displaystyle\int_0^2 \sqrt{4 - x^2}\cdot \operatorname{sgn}(x-1)\,\mathrm{d}x = \pi - \sqrt{3}$

結論. $\displaystyle\sum_{k=1}^{m} k^2 = \frac{m(m+1)(2m+1)}{6}$。

例. 求 $\displaystyle\int_0^1 x^2\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}: \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}\right\}$，則 $\Delta x_k = \frac{1}{n}\ \forall\, k = 1, 2, \ldots, n$ 且 $\|P\| \to 0$ 當 $n \to \infty$。$f(x) = x^2$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} u_k\,\Delta x_k = \sum_{k=1}^{n} \frac{k^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2$，$L(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} l_k\,\Delta x_k = \sum_{k=1}^{n} \frac{(k-1)^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n-1} k^2$，$L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。由 $\displaystyle\sum_{k=1}^{m} k^2 = \frac{m(m+1)(2m+1)}{6}$，$\displaystyle\lim_{n\to\infty} U(f,\mathbb{P}) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n} k^2 = \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3}$，$\displaystyle\lim_{n\to\infty} L(f,\mathbb{P}) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n-1} k^2 = \lim_{n\to\infty} \frac{(n-1)n(2n-1)}{6n^3} = \frac{1}{3}$，故由夾擊定理 $\displaystyle\lim_{n\to\infty} R(f,\mathbb{P}) = \frac{1}{3}$。

結論. 若 $r \in \mathbb{R}$，$\displaystyle\sum_{k=0}^{n} r^k = \frac{r^{n+1} - 1}{r - 1}$。

例. 求 $\displaystyle\int_0^1 e^x\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}: \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}\right\}$，則 $\Delta x_k = \frac{1}{n}\ \forall\, k = 1, 2, \ldots, n$ 且 $\|P\| \to 0$ 當 $n \to \infty$。$f(x) = e^x$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} u_k\,\Delta x_k = \sum_{k=1}^{n} e^{\frac{k}{n}}\cdot\frac{1}{n} = \frac{1}{n}\frac{e^{\frac{1}{n}}\left(e^{\frac{n}{n}} - 1\right)}{e^{\frac{1}{n}} - 1} = \frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}} - 1}$，$L(f,\mathbb{P}) = \displaystyle\sum_{k=1}^{n} l_k\,\Delta x_k = \sum_{k=1}^{n} e^{\frac{k-1}{n}}\cdot\frac{1}{n} = \frac{1}{n}\frac{e^{0}\left(e^{\frac{n}{n}} - 1\right)}{e^{\frac{1}{n}} - 1} = \frac{e-1}{n}\frac{1}{e^{\frac{1}{n}} - 1}$，$L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。$\displaystyle\lim_{n\to\infty} U(f,\mathbb{P}) = \lim_{n\to\infty} \frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}} - 1} = (e-1)\lim_{x\to 0+} \frac{xe^x}{e^x - 1} = (e-1)\lim_{x\to 0+} \frac{e^x + xe^x}{e^x} = e - 1$，$\displaystyle\lim_{n\to\infty} L(f,\mathbb{P}) = \lim_{n\to\infty} \frac{e-1}{n}\frac{1}{e^{\frac{1}{n}} - 1} = (e-1)\lim_{x\to 0+} \frac{x}{e^x - 1} = (e-1)\lim_{x\to 0+} \frac{1}{e^x} = e - 1$，故由夾擊定理 $\displaystyle\lim_{n\to\infty} R(f,\mathbb{P}) = e - 1$。

定理 (微積分基本定理（Fundamental Theorem of Calculus，FTC）).

1. 若 f 在 $[a,b]$ 連續，令 $F(x)=\int_a^x f(\tau)\,\mathrm{d}\tau$ 且 $a\leqslant x\leqslant b$，則 $F'(x)=f(x)\ \forall\, x\in[a,b]$。
2. 若 $G'(x)=f(x)\ \forall\, x\in[a,b]$，則 $\int_a^b f(x)\,\mathrm{d}x=G(b)-G(a)\equiv G(x)\Big|_a^b$。

註. 由 FTC，$\int_a^b f(x)\,\mathrm{d}x$ 可由 f 的反導函數（不定積分）得出，不需繁複極限計算！

證.

1. WLOG 令 $h>0$。則 $\left|\dfrac{F(x+h)-F(x)}{h}-f(x)\right|=\left|\dfrac{1}{h}\int_x^{x+h}(f(\tau)-f(x))\,\mathrm{d}\tau\right|\leqslant\dfrac{1}{h}\int_x^{x+h}\big|f(\tau)-f(x)\big|\,\mathrm{d}\tau\leqslant\max\limits_{x\leqslant\tau\leqslant x+h}\big|f(\tau)-f(x)\big|\to 0$，當 $h\to 0$ 且 f 為連續。
2. $(G(x)-F(x))'=f(x)-f(x)=0\ \forall\, x\in[a,b]$，故 $G(x)-F(x)$ 在 $[a,b]$ 為常數 $\implies G(a)-F(a)=G(b)-F(b)\implies G(b)-G(a)=F(b)-F(a)=\int_a^b f(x)\,\mathrm{d}x$。

例 (由 FTC 求定積分).

1. x^2 之反導函數為 $\dfrac{x^3}{3}$，故 $\int_0^1 x^2\,\mathrm{d}x=\dfrac{1^3}{3}-\dfrac{0^3}{3}=\dfrac{1}{3}$。
2. e^x 之反導函數為 e^x，故 $\int_0^1 e^x\,\mathrm{d}x=e^1-e^0=e-1$。

結論. $\displaystyle\sum_{k=1}^n\cos kx=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$

證. 考慮 $\displaystyle\sum_{k=1}^n e^{ikx}=\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}$,則 $\displaystyle\sum_{k=1}^n\cos kx=\Re\Big\{\sum_{k=1}^n e^{ikx}\Big\}=\Re\Big\{\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}\Big\}=\Re\Big\{\frac{e^{ix}\,e^{\frac{inx}{2}}\left(e^{-\frac{inx}{2}}-e^{\frac{inx}{2}}\right)}{e^{\frac{ix}{2}}\left(e^{-\frac{ix}{2}}-e^{\frac{ix}{2}}\right)}\Big\}=$
$\displaystyle\frac{\cos\frac{(n+1)x}{2}\,\sin\frac{nx}{2}}{\sin\frac{x}{2}}=\frac{\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)+\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)}{2\,\sin\frac{x}{2}}=\frac{\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}}{2\,\sin\frac{x}{2}}=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$。

例. 求 $\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x$。

解.

- (極限定義) 建立 $\left[0,\frac{\pi}{2}\right]$ 分割 $\mathbb{P}:\left\{0,\frac{\pi}{2n},\frac{2\pi}{2n},\ldots,\frac{n\pi}{2n}\right\}$，則 $\Delta x_k=\dfrac{\pi}{2n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$,則積分為 $\displaystyle\lim_{n\to\infty}R(f,\mathbb{P})=\sum_{k=1}^n f(\xi_k)\,\Delta x_k=\lim_{n\to\infty}\sum_{k=1}^n\cos\frac{k\pi}{2n}\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\sum_{k=1}^n\cos\left(k\left(\frac{\pi}{2n}\right)\right)\cdot\frac{\pi}{2n}=$ $\displaystyle\lim_{n\to\infty}\frac{1}{2}\left(\frac{\sin\left(\left(n+\frac{1}{2}\right)\cdot\frac{\pi}{2n}\right)}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}-1\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{\sin\left(\left(n+\frac{1}{2}\right)\frac{\pi}{2n}\right)\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\lim_{n\to\infty}\sin\left(\frac{1}{2}+\frac{1}{4n}\right)\pi\cdot\lim_{n\to\infty}\frac{\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=$ $\displaystyle\sin\frac{\pi}{2}\cdot 1=1.$
- (FTC) $\cos x$ 之反導函數為 $\sin x$，故 $\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x=\sin\dfrac{\pi}{2}-\sin 0=1$。

性質. 令 $F(x)=\int_{v(x)}^{u(x)}f(\tau)\,\mathrm{d}\tau$，則 $F'(x)=f(u(x))\cdot u'(x)-f(v(x))\cdot v'(x)$。

證. 令 $a \in \mathbb{R}$, $F(x) = \int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau$。令 $G(x) \equiv \int_a^x f(\tau)\,\mathrm{d}\tau$, 則 $G'(x) = f(x)$, $F(x) = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau = G(u(x)) - G(v(x))$；故 $F'(x) = (G(u(x)) - G(v(x)))' = G'(u(x)) \cdot u'(x) - G'(v(x)) \cdot v'(x) = f(u(x)) \cdot u'(x) - f(v(x)) \cdot v'(x)$。

例.

1. $F(x) = \int_1^x \frac{1}{1+\tau^4}\,\mathrm{d}\tau \implies F'(x) = \frac{1}{1+x^4}$
2. $F(x) = \int_2^{\sqrt{x}} \sin\tau\,\mathrm{d}\tau \implies F'(x) = \sin\sqrt{x} \cdot \frac{1}{2\sqrt{x}}$
3. $F(x) = \int_x^{2x} \tau^3\,\mathrm{d}\tau \implies F'(x) = (2x)^3 \cdot 2 - x^3 \cdot 1 = 15x^3$

定積分變數變換

結論.

- 先求得反導函數後再代入：$\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = f(g(x))\Big|_{x=a}^{x=b} = f(g(b)) - f(g(a))$
- 變數變換並改變積分範圍：$\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = \int_a^b f'(g(x))\,\mathrm{d}g(x) = \int_{g(a)}^{g(b)} f'(u)\,\mathrm{d}u = f(u)\Big|_{u=g(a)}^{u=g(b)} = f(g(b)) - f(g(a))$

例. 求 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入：令 $u = 1+x^4$，則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \frac{\mathrm{d}u}{4}$，故 $\int x^3(1+x^4)^3\,\mathrm{d}x = \int (1+x^4)^3\,x^3\,\mathrm{d}x = \int u^3\,\frac{\mathrm{d}u}{4} = \frac{u^4}{16} + c = \frac{(1+x^4)^4}{16} + c$。故 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \frac{(1+x^4)^4}{16}\Big|_{x=0}^{x=1} = \frac{(1+1^4)^4 - (1+0^4)^4}{16} = \frac{15}{16}$。
- 變數變換並改變積分範圍：令 $u = 1+x^4$，則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \frac{\mathrm{d}u}{4}$。積分範圍 x 由 0 至 1，則變數變換後 u 由 $1+0^4 = 1$ 至 $1+1^4 = 2$，故 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \int_0^1 (1+x^4)^3\,x^3\,\mathrm{d}x = \int_1^2 u^3\,\frac{\mathrm{d}u}{4} = \frac{1}{4}\int_1^2 u^3\,\mathrm{d}u = \frac{1}{16}u^4\Big|_{u=1}^{u=2} = \frac{2^4 - 1^4}{16} = \frac{15}{16}$。

例. 求 $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入：令 $u = 1+x^2$，則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$，故 $\int \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u} + c = 4\sqrt{1+x^2} + c$。故 $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = 4\sqrt{1+x^2}\Big|_{x=0}^{x=\sqrt{3}} = 4\sqrt{1+3} - 4\sqrt{1} = 4$。

- 變數變換並改變積分範圍：令 $u = 1 + x^2$，則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$。積分範圍 x 由 0 至 $\sqrt{3}$，則變數變換後 u 由 $1 + 0^2 = 1$ 至 $1 + (\sqrt{3})^2 = 4$，故 $\displaystyle\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int_1^4 \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u}\Big|_{u=1}^{u=4} = 4\,(\sqrt{4} - \sqrt{1}) = 4$。

例. 求 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入:令 $u = \cos x$,則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$,故 $\displaystyle\int 3\cos^2 x \sin x\,\mathrm{d}x = -3\int u^2\,\mathrm{d}u = -u^3 + c = -\cos^3 x + c$。故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\cos^3 x\Big|_{x=0}^{x=\pi} = -(\cos^3\pi - \cos^3 0) = -((-1)^3 - 1^3) = 2$。
- 變數變換並改變積分範圍：令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$。積分範圍 x 由 0 至 π，則變數變換後 u 由 $\cos 0 = 1$ 至 $\cos\pi = -1$，故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\int_1^{-1} 3u^2\,\mathrm{d}u = -u^3\Big|_{u=1}^{u=-1} = -((-1)^3 - 1^3) = 2$。

4.3 積分技巧

三角函數代換

部份分式

4.4 瑕積分

定義 (瑕積分).

- 若 (i) f 在 I 有無限函數值或 (ii) I 為無限區間，$\displaystyle\int_I f(x)\,\mathrm{d}x$ 稱為瑕積分 (improper integral)。
- $\displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \lim_{\substack{a\to-\infty \\ b\to\infty}} \int_a^b f(x)\,\mathrm{d}x$。

例. 求 $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x$。

解. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x = 2\int_0^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim_{b\to\infty}\tan^{-1} b = 2\cdot\frac{\pi}{2} = \pi$。

例. 求 $\displaystyle\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x$。

解. $\displaystyle\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,\mathrm{d}x = \lim_{b\to\infty}\big(-\frac{1}{b} + 1\big) = 1$

例. 求 $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x$。

解. $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{c\to 0+}\int_c^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{c\to 0+} 2\sqrt{x}\Big|_c^1 = \lim_{c\to 0+}(2 - 2\sqrt{c}) = 2$

結論. 給定 $0 < a < \infty$。

- $\displaystyle\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x$ 當 $p>1$ 收斂至 $\dfrac{a^{1-p}}{p-1}$，當 $p\leqslant 1$ 發散至 ∞。
- $\displaystyle\int_0^a \frac{1}{x^p}\,\mathrm{d}x$ 當 $p<1$ 收斂至 $\dfrac{a^{1-p}}{1-p}$，當 $p\geqslant 1$ 發散至 ∞。

結論. 令 $-\infty\leqslant a<b\leqslant\infty$，函數 f、g 在 (a,b) 連續，$0\leqslant f(x)\leqslant g(x)\ \forall x$。

- 若 $\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 收斂，$\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 收斂。
- 若 $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 發散，$\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 發散。

例. 證明 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

解. 由 $e^{-x^2}\leqslant 1\ \forall 0\leqslant x<1$ 及 $e^{-x^2}\leqslant e^{-x}\ \forall x\geqslant 1$，$\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x=\int_0^1 e^{-x^2}\,\mathrm{d}x+\int_1^\infty e^{-x^2}\,\mathrm{d}x\leqslant\int_0^1 1\,\mathrm{d}x+\int_1^\infty e^{-x}\,\mathrm{d}x=1+\frac{1}{e}$，故 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

例. 證明 $\forall x>0$，$\displaystyle\Gamma(x)\equiv\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

解. 由 $\displaystyle\lim_{t\to\infty}t^{x-1}e^{-\frac{t}{2}}=0,\exists T>0$ 使 $t^{x-1}e^{-\frac{t}{2}}\leqslant 1\ \forall t\geqslant T$。則 $\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t=\int_0^T t^{x-1}e^{-t}\,\mathrm{d}t+\int_T^\infty t^{x-1}e^{-t}\,\mathrm{d}t\leqslant\int_0^T t^{x-1}\,\mathrm{d}t+\int_T^\infty 1\cdot e^{-\frac{t}{2}}\,\mathrm{d}t=\frac{T^x}{x}+2e^{-\frac{T}{2}}$。故 $\forall x>0$；$\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

例. 證明 $\forall x>0$，$\Gamma(x+1)=x\,\Gamma(x)$。

解. $\displaystyle\Gamma(x+1)=\int_0^\infty t^xe^{-t}\,\mathrm{d}t=\lim_{\substack{a\to 0+\\ b\to\infty}}\int_a^b t^xe^{-t}\,\mathrm{d}t$。令 $u=t^x$，則 $\mathrm{d}u=xt^{x-1}\,\mathrm{d}t$。令 $\mathrm{d}v=e^{-t}\,\mathrm{d}t$，則 $v=-e^{-t}$。故 $\displaystyle\lim_{\substack{a\to 0+\\ b\to\infty}}\int_a^b t^xe^{-t}\,\mathrm{d}t=\lim_{\substack{a\to 0+\\ b\to\infty}}\Big(-t^xe^{-t}\Big|_a^b+\int_a^b e^{-t}\cdot xt^{x-1}\,\mathrm{d}t\Big)=0+x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t=x\,\Gamma(x)$。

例. 由 $\displaystyle\int_0^\infty e^{-t^2}\,\mathrm{d}t=\frac{\sqrt{\pi}}{2}$，求 $\Gamma\Big(\dfrac{1}{2}\Big)$ 與 $\Gamma\Big(\dfrac{3}{2}\Big)$。

解.

1. $\displaystyle\Gamma\Big(\frac{1}{2}\Big)=\int_0^\infty t^{-\frac{1}{2}}e^{-t}\,\mathrm{d}t=\int_0^\infty\frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t$。令 $u=\sqrt{t}$，則 $t=u^2$，$\mathrm{d}u=\dfrac{1}{2\sqrt{t}}\,\mathrm{d}t\implies\dfrac{1}{\sqrt{t}}\,\mathrm{d}t=2\,\mathrm{d}u$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\sqrt{0}=0$ 至 $\sqrt{\infty}=\infty$，故 $\displaystyle\Gamma\Big(\frac{1}{2}\Big)=\int_0^\infty\frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t=2\int_0^\infty e^{-u^2}\,\mathrm{d}u=2\cdot\frac{\sqrt{\pi}}{2}=\sqrt{\pi}$。
2. 由 $\Gamma(x+1)=x\,\Gamma(x)$，$\Gamma\Big(\dfrac{3}{2}\Big)=\Gamma\Big(\dfrac{1}{2}+1\Big)=\dfrac{1}{2}\,\Gamma\Big(\dfrac{1}{2}\Big)=\dfrac{1}{2}\sqrt{\pi}$。

例. 由 $\displaystyle\int_0^\infty e^{-t^2}\,\mathrm{d}t=\frac{\sqrt{\pi}}{2}$ 且 $\alpha>0$，求 $\displaystyle\int_0^\infty t\,e^{-\alpha\,t^2}\,\mathrm{d}t$ 與 $\displaystyle\int_0^\infty t^2\,e^{-\alpha\,t^2}\,\mathrm{d}t$。

解.

1. 令 $u=\alpha\,t^2$，則 $\mathrm{d}u=2\,\alpha\,t\,\mathrm{d}t \implies t\,\mathrm{d}t=\dfrac{\mathrm{d}u}{2\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2=0$ 至 $\alpha\cdot\infty^2=\infty$，故 $\displaystyle\int_0^\infty t\,e^{-\alpha\,t^2}\,\mathrm{d}t=\int_0^\infty e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha}=\frac{1}{2\,\alpha}\int_0^\infty e^{-u}\,\mathrm{d}u=\frac{1}{2\,\alpha}\Big(-e^{-u}\Big|_0^\infty\Big)=\frac{1}{2\,\alpha}$。

2. 令 $u=\alpha\,t^2$，則 $t=\dfrac{\sqrt{u}}{\sqrt{\alpha}}$，$\mathrm{d}u=2\,\alpha\,t\,\mathrm{d}t \implies t\,\mathrm{d}t=\dfrac{\mathrm{d}u}{2\,\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2=0$ 至 $\alpha\cdot\infty^2=\infty$，故 $\displaystyle\int_0^\infty t^2\,e^{-\alpha\,t^2}\,\mathrm{d}t=\int_0^\infty t\,e^{-\alpha\,t^2}\,t\,\mathrm{d}t=\int_0^\infty \frac{\sqrt{u}}{\sqrt{\alpha}}\,e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha}=\frac{1}{2\,\alpha^{\frac{3}{2}}}\int_0^\infty u^{\frac{1}{2}}\,e^{-u}\,\mathrm{d}u=\frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\Gamma\Big(\frac{3}{2}\Big)=\frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\frac{1}{2}\sqrt{\pi}=\frac{\sqrt{\pi}}{4\,\alpha^{\frac{3}{2}}}$。